Tony Hsu Tai

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Education

Northeastern University Sep 2024 - May 2027
M.S. in Computer Science | ***GPA: 3.92/4.00*** *Boston, MA*
- Coursework: Data Structures and Algorithms, Foundations of AI, Scalable Distributed Systems, Database Management Systems

Northeastern University Sep 2016 - May 2021
Accelerated B.S./M.S. Program in Industrial Engineering | ***GPA: 3.75/4.00*** *Boston, MA*

Professional Experience

Software Developer Intern May 2026 – Present
Sentinel Group *Wakefield, MA*
- Built an **Airflow ETL pipeline** and **NestJS REST services** that migrates payroll and deposit configurations from an Infor CRM and Oracle system into the new payroll platform, replacing manual data entry to a one-click migration with rollback across 3,000+ clients.
- Enabled **dynamic chart-based PDF reports** through the existing document-generation workflow, letting template authors add **data-bound charts** that render per document at generation time, with no performance degradation for current chart-less document generation.
- Implemented an **event-driven client notification pipeline using Amazon SQS** with dead-letter handling, content-based de-duplication, and persisted timestamps to prevent duplicate delivery under retries, giving clients automatic confirmation for every SFTP file upload.
- Developed a **Kubernetes-based Azure DevOps to Glean repository crawler** that tracks repositories tip commit SHA to skip unchanged code and syncs permission-aware visibility, giving our enterprise search platform full visibility into our code patterns across 25,000+ files.
- Overhauled the local development stack by replacing LocalStack with Floci for SQS/S3 emulation and **automating environment setup with a single setup script** for 12 service configurations and AWS resources, cutting setup time from 2 hours to under 15 minutes.

Operations Manager Sep 2021 – Jun 2024
Boston Microgreens *Boston, MA*
- Built an **Excel-based scheduling system** that automated seeding, germination, and harvest planning for 50+ weekly orders, becoming the farm's single source of truth for reliable crop and order data that remains in daily use after 4+ years.
- Designed a **germination tracking system using relational databases** and computed fields to track crop stage transitions, automatically flagging what crops to check and move to irrigation area, reducing task time by 63%.
- Built a **per-crop cost model** that combined each variety's raw material costs with operation costs allocated by growth time and resource usage, revealing 11% of varieties were not generating profit and driving strategic adjustments to improve margins.

Personal Projects

AI Financial Analysis Platform | *TypeScript, Node.js, PostgreSQL, React, Gemini API* Dec 2025 – June 2026
- Built an **asynchronous PDF statement ingestion pipeline** in TypeScript and PostgreSQL that extracts account and transaction data with layered regex and tracks each upload through a status lifecycle.
- Engineered a **three-tier transaction classifier** using 890+ merchant aliases, 55 regex rules, and a **Gemini fallback** that sends only unmatched transactions in one batched call per statement.
- Designed an **LLM finance chat** that turns questions into **user-scoped, parameterized SQL** through schema-aware prompting and fills answer templates by querying data, so all figures come from the database.

Agricultural Operations System | *Python, TypeScript, MySQL, React, FastAPI* Nov 2025 – Dec 2025
- Designed a full-stack farm operations platform with React, FastAPI and MySQL, **replacing spreadsheet workflows with a relational data model** that enforces integrity through foreign-key constraints and stored-procedure validation.
- Built a **SQL view and stored procedures that work backward from customer delivery dates** to compute planting, germination dates, irrigation frequency and trays counts, fully automating daily production planning.
- Implemented **JWT authentication** with **bcrypt** password hashing and expiring signed tokens, linking user accounts to employee records so requests resolve the acting employee and attribute orders to them.

Pokémon Battle Game | *TypeScript, React, Socket.IO, Express.js, Node.js* Jul 2025 – Sep 2025
- Built a **server-authoritative real-time multiplayer system** with Socket.IO and Express, using room-code lobbies and a typed event protocol that holds each turn until both players submit, then sends each client its own view of the battle state.
- Saved each server-resolved turn into an **ordered event log** (attack, switch, faint) that the React client plays back from a queue, driving Motion animations for attacks, switches, HP changes and faints so both players see the same sequence.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C/C++, SQL, HTML/CSS

Frameworks & Libraries: React, Node.js, Nest.JS, Next.js, Express.js, FastAPI, Tailwind CSS

Database Management: PostgreSQL, MySQL, MongoDB

Tools & Platforms: AWS (S3, EKS, SQS), Git/Github, Azure DevOps, Docker, Postman, Claude Code

Achievements

2026 **Winner - Best Game**, GoatHacks 2026 - Worcester Polytechnic Institute *Worcester, MA*